National College of Ireland

MSc in Cloud Computing, Year 1, MSCCLOUD1JAN26I
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Release Date: Tuesday 3rd February 2026
Submission Deadline: Wednesday 1st of April 2026 at 16:00

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Cloud DevOpsSec
Continuous Assessment (CA): Project

Weight: The assignment will be marked out of 100. This CA represents 60% of the overall marks for this module.

Instructions: This is an individual assessment/project. You are required to submit all the project deliverables via Moodle using the dedicated submission pages.

SUBMISSION DETAILS: All the project deliverables must be submitted via Moodle before the deadline using the dedicated submission pages. **Late submissions will be penalised** unless an extension is applied for through NCI360 and it is approved. The report deliverable should be formed from paragraphs and should NOT contain ordered and/or unordered (e.g., bullet points) lists.

Academic Integrity: IMPORTANT: It is your responsibility to avoid plagiarism. Please read the comprehensive guidelines on academic honesty and academic integrity, and how to avoid plagiarism, made available by the NCI Library (https://libguides.ncirl.ie/referencingandavoidingplagiarism).
It is expected that the Submitted Assignment is FULLY THE WORK OF THE STUDENT except where quoted material is clearly indicated. The use of Artificial Intelligence (AI) text generators (e.g., ChatGPT, Quillbot, GitHub Copilot, etc.) to rewrite/paraphrase text and/or to generate code could be considered a breach of academic integrity, as you are not doing the work yourself and are using AI to carry out the work YOU should be doing. Paraphrasing is a very important part of an assignment as it proves a person understood the information enough to put it into their own words.

NOTE: YOU ARE NOT ALLOWED TO PUBLISH THIS ASSIGNMENT BRIEF OR A PART THEREOF ON ANY WEBSITES. YOU ARE NOT ALLOWED TO PUBLISH/SHARE YOUR SOLUTION WITH OTHERS. All work submitted should be YOUR own. Conferring with others is NOT permitted. This is NOT a collaborative assessment.

TURNITIN: All submissions will be electronically screened for evidence of academic misconduct, e.g., plagiarism, collusion, AI generated text and/or code, misrepresentation, etc. Any submission showing evidence of academic misconduct will be investigated in accordance with the provisions of the Academic Integrity Policy.

Introduction

The assessment of the Cloud DevOpsSec module is based on two assessment components:

a) *Project* which represents 60% of the module assessment, and
b) *Terminal Exam* which represents 40% of the module assessment.

The learning outcomes of the Cloud DevOpsSec module are as follows:

LO1. Critically analyse different techniques to perform code analysis, plan and implement static code analysis.
LO2. Develop and implement a plan for provisioning and configuration of software applications through CI/CD.
LO3. Critically evaluate and implement methodologies for secure application development and secure execution in production.

The Project assesses LO1, LO2, and LO3.

Project Description

For this assignment, you are required to develop a dynamic cloud-based application. You are required to build, deploy, and update your web application by setting up a full lifecycle CI/CD pipeline. Your application must be deployed and hosted on a *public cloud provider*.

Your dynamic web-based application should:

- Accept input from the user and validate that input
- Provide CRUD functionalities
- Use a suitable data storage solution (i.e., based on the requirements of your application)
- Deploy your application to a suitable public cloud platform. The deployed application must not be modified after the submission deadline. The examiner should be able to view your deployed application without having to register for any account with the public cloud provider where you deployed your application (i.e., the application **(not its source code!)** should be publicly accessible via a URL). This publicly accessible URL should not be shared with anyone else but the examiners of this instance of the Cloud DevOpsSec module at NCI.

You must conduct some independent research and include any relevant bibliography in the accompanying report.

Please refrain from sharing or reusing code from other students and/or external stakeholders unless otherwise instructed.

On completion, you will document the process and reflect on it through the deliverables listed in the next section.

Project Deliverables

You are required to document the process of developing the cloud-based application and the CI/CD pipeline setup, and reflect on it through the following deliverables:

1. A **project report** (7-8 pages formatted using the IEEE Conference double-column template[1]) which should include:
    - NCI Project Submission Sheet/ Project Cover Sheet
    - Headline: title of the report, your name, student number, module, programme, and date
    - Abstract – a 150-250-word executive summary of the project and the main results
    - Section 1: Introduction – motivation for your project and its main objectives. A brief description of your application.
    - Section 2: Continuous integration, continuous delivery and deployment of your application
        - Document the CI/CD pipeline including a diagram of the entire workflow; the diagram should indicate where the different tools/cloud-based services are used to support the CI/CD pipeline, and the relationships between these services/tools. Ensure that you distinguish between the CI and CD in the diagram. The diagram should

[1] https://www.overleaf.com/read/bxnmxqvgkcnb

be fully explained in text. Note that the diagram should be created by you based on the pipeline and stages you set up for your project.

- Include the URL to your deployed application
- Document the CI/CD pipeline in action – document how a change in the source code of your application flows through the pipeline. Note that the change must be done in the source code, in the development environment, via the integrated development environment you used to develop your application. Ensure that you include relevant screenshots of the pipeline you set up for your project that capture that code change flow, together with all the commands used.

Note: At all times you **must use** a **private repository** for versioning control (e.g., GitHub, Bitbucket, etc.)

- Section 3: Critically analyse and document the approach you took for performing static code analysis, including security vulnerabilities analysis. Document your findings from performing static code analysis and security vulnerabilities analysis, also document their fixes. Ensure that you include screenshots of the findings from performing static code analysis and security vulnerability analysis on your application.
- Section 4: Conclusions including findings/interpretations – what did you learn and find out? Include a short reflection on developing this project. If you were to implement this project again, what would you do differently?
- Section 5: References – a complete list of academic works and/or online materials used in the project. References should be included as in-text citations using the IEEE referencing style. Note that a good starting point to find academic works is the NCI Library Guide on Cloud Computing at https://libguides.ncirl.ie/cloudcomputing

Note: Ensure that you include in the report for all the previously mentioned elements demonstration snippets and/or screenshots of the commands and tools used and screenshots of the findings from static code analysis AND security vulnerability analysis, where appropriate.

IMPORTANT: Anything after Page 8 will not be considered, i.e., it WILL NOT BE MARKED! (Note that the NCI Project Submission Sheet/ Project Cover Sheet is not considered as part of the page count.)

2. The **source code artefacts** submission (packed as a compressed ZIP file) should include:
   - Source code of the solution (includes commented source code of the application together with any scripts used for automation and configuration)

     Note: Ensure that you include substantial meaningful comments in YOUR source code to document your ORIGINAL contributions.

3. **Video of project presentation, project demonstration and answers to questions. A maximum 10-minute video submission that should include the following:**
   - A concise (approximate 1-2 minutes) presentation of the motivation and high-level description of the idea of the project.
   - Demonstration – give a demonstration of your application highlighting the main features; give a demonstration of the CI/CD pipeline by making a change in one of the features of your application (i.e., modify its source code in the development environment via the integrated development environment (IDE) you used to develop your application) and show how the change flows through the CI/CD pipeline and describe the operations performed in each stage of the pipeline, including demonstrating the modified feature in the deployed application. **IMPORTANT**: **DO NOT**! **edit the source code directly in the GitHub (i.e., edit the source code via the IDE you used) as a demonstration of such a pipeline WILL NOT be considered/ WILL NOT BE MARKED!**
   - Answers to questions/items which you will receive in advance from your lecturer

IMPORTANT: Every second after the 10-minute video will not be considered, i.e., it WILL NOT BE MARKED! Please record your project video and upload it to YouTube as an UNLISTED video (for instructions, please consult https://support.google.com/youtube/answer/157177).

NOTE: The examiners reserve the right to conduct live mini presentations with a sample of the students, where students will demonstrate their project including the CI/CD pipeline, and will provide answers to questions related to their project. Also, students may be required to undergo a viva (oral examination) if there is suspicion about the validity of their submitted work.

The YouTube URL with your project presentation, demonstration and answers to questions video must be included in your report (e.g. https://youtu.be/mxT233EdY5c) AND must be publicly accessible.
Please test your URL well before your submission using a private window of your browser.

Assessment Criteria

The Project will be assessed based on the assessment criteria shown in Table 1 and the marking rubric shown in Table 2.

Table 1 Assessment Criteria

The CI/CD pipeline	30%
Static code analysis, including security vulnerability analysis	35%
Conclusions and findings & References	10%
Video Presentation & Demonstration & Answers to Questions	25%

Table 2 Marking Rubric

Grade Criterion	H1 (> 70%)	60-69%	50-59%	40-49%	Fail (< 40%)
The CI/CD pipeline: 30%	Excellent/very good application of CI/CD pipeline design and development in terms of appropriate methodology and tools/cloud-based services. Excellent/very good discussion and documentation of the CI/CD pipeline. Excellent/very good demonstration of implemented dynamic input validation in the deployed application.	Good application of CI/CD pipeline design and development in terms of appropriate methodology and tools/cloud-based services. Good discussion and documentation of the CI/CD pipeline. Good demonstration of implemented dynamic input validation in the deployed application.	Adequate application of CI/CD pipeline design and development in terms of appropriate methodology and tools/cloud-based services. Adequate discussion and documentation of the CI/CD pipeline. Adequate demonstration of implemented dynamic input validation in the deployed application.	Weak application of CI/CD pipeline design and development in terms of appropriate methodology and tools/cloud-based services. Weak discussion and documentation of the CI/CD pipeline. Weak demonstration of implemented dynamic input validation in the deployed application.	Poor application of CI/CD pipeline design and development in terms of appropriate methodology and tools/cloud-based services. Poor discussion and documentation of the CI/CD pipeline. Poor/Inexistent demonstration of implemented dynamic input validation in the deployed application.
Static code analysis, including security vulnerability analysis: 35%	Excellent/very good critical analysis of substantive and relevant findings that incorporate both findings from static analysis and security vulnerability analysis, together with their fixes/solutions.	Good critical analysis of substantive and relevant findings that incorporate both findings from static analysis and security vulnerability analysis. together with their fixes/solutions.	Adequate critical analysis of relevant findings that incorporate findings from static analysis and/or security vulnerability analysis, together with their fixes/solutions.	Limited critical analysis of relevant findings that incorporate findings from static analysis and/or security vulnerability analysis. Limited documentation of how those findings were fixed/solved.	Very limited and poor/inexistent critical analysis of relevant findings that incorporate findings from static analysis and/or security vulnerability analysis. Very limited and/or inexistent documentation of how those findings were fixed/solved.
Conclusions and Findings & References: 10%	Excellent/very good conclusions, insightful findings and reflection. References are complete, consistent, appropriately, and correctly used.	Good conclusions and good discussion of findings and reflection. Most references are complete, consistent, appropriately and correctly used.	Adequate conclusions and adequate discussion of findings and reflection. Adequate, consistent and appropriate referencing.	Limited/weak conclusions and limited/weak discussion of findings and reflection. References are few and/or mostly incomplete and/or inconsistent and/or inappropriately used.	Very limited and poor/inexistent conclusions. Very limited and poor/inexistent discussion of findings and reflection. References (if any) are incomplete and/or inconsistent and/or inappropriately used.
Video Presentation & Demonstration & Answers to Questions: 25%	Excellent well directed presentation and demonstration with impeccable handling of questions.	Clear presentation and demonstration with good handling of questions.	Neat oral presentation and demonstration and acceptable handling of questions.	Poor oral presentation and demonstration and weak handling of questions.	Unacceptable oral presentation and demonstration and poor handling of questions.

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